

Paper Review

Alias-Free Generative Adversarial Networks (StyleGAN3)

Reviewed paper T. Karras, M. Aittala, S. Laine, E. Härkönen, J. Hellsten, J. Lehtinen, T. Aila. *Alias-Free Generative Adversarial Networks*. NeurIPS 2021 (arXiv:2106.12423).

Venue rank NeurIPS, A* (CORE).

Reviewer Shamima Hossain, ID 21141119

Course CSE 754: Deep Learning for Computer Vision

Date September 19, 2026

Summary. StyleGAN3 makes a diagnosis and a repair. The diagnosis is that GAN generators secretly depend on absolute pixel coordinates, so fine detail sticks to the image grid instead of to the surfaces it belongs to; the paper traces this to *aliasing* introduced by the generator’s own upsampling and nonlinearities. The repair is to reinterpret every feature map as a bandlimited continuous signal and to make each layer respect that interpretation. The outcome is a generator that is translation- and rotation-equivariant to sub-pixel precision while matching StyleGAN2’s FID. This review argues that the diagnosis is the paper’s real contribution, that the evaluation is narrower than the claims, and that the most interesting open question (whether equivariance is even desirable outside natural images) is left untouched.

1 Problem Statement

Generative adversarial networks [3] build images hierarchically: coarse, low-resolution features are upsampled, mixed by convolutions, and enriched with new detail at each scale. The implicit promise of that design is that the position of a fine feature is inherited from the coarse feature it belongs to: move the head and the nose moves with it, and the skin pores move with the nose.

The paper observes that this promise is not kept. In StyleGAN2 [7], much of the fine detail is instead pinned to *absolute pixel coordinates*. The authors name the symptom “**texture sticking**”: during a latent-space interpolation, hair and fur do not travel with the head but remain fixed to the screen, producing horizontal streaks when a pixel column is stacked over time (the paper’s Figure 1). Averaging images generated from a small latent neighbourhood should give a uniformly blurry result; with StyleGAN2 some regions stay unnaturally sharp, because they never moved.

The paper’s central claim is that this is not a training artefact or a capacity problem but a **signal-processing bug**. Generators can bypass the intended hierarchy by reading positional information that leaks in through four side channels: image borders, per-pixel noise inputs, positional encodings, and the one the paper focuses on, **aliasing**. Whenever a layer produces frequencies above the Nyquist limit of its own sampling grid, those frequencies fold back as low-frequency content that is tied to the grid rather than to the image. The generator can then use this folded content as a coordinate reference.

Significance. A still image whose texture is glued to the grid may look perfectly good; the defect only becomes visible in motion. The problem is therefore invisible to FID [4], the field’s dominant metric, which is exactly why it persisted through several generations of state-of-the-art models. The paper’s stated goal is a generator whose sub-pixel feature positions are inherited

entirely from coarse features, making GANs usable for video and animation rather than for isolated stills.

Assessment. The problem statement is unusually strong. It identifies a defect that the community’s standard metric structurally cannot see, gives it a falsifiable mechanistic cause, and motivates it with a demonstration that needs no metric at all. The weakness (developed in Section 4) is that the motivating example is about *local, hierarchical* motion (nose with head), whereas what the paper ultimately delivers and measures is equivariance to *global* rigid transforms of the whole image.

2 Methodology and Analysis

2.1 Continuous-signal formulation

The paper’s method follows from a single reframing. A feature map is not an array of numbers but a *sampled representation of a bandlimited continuous signal*. By the Nyquist–Shannon theorem, a feature map sampled on a grid of spacing $1/s$ can faithfully represent only frequencies below $s/2$, and the discrete array Z and its continuous counterpart z are related by convolution with an ideal interpolation filter,

$$z(x) = (\phi_s * Z)(x), \quad \phi_s(x) = \text{sinc}(sx_0) \cdot \text{sinc}(sx_1). \quad (1)$$

Every operation in the generator must then be understood as an operation on the *continuous* signal. An operation is admissible only if (i) it commutes with translation and rotation in the continuous domain, and (ii) its output remains bandlimited to the grid that will store it. Any layer violating (ii) aliases, and aliasing is what leaks coordinates.

This reframing immediately isolates the culprit. Convolutions and upsampling are linear and easy to make well-behaved. **Pointwise nonlinearities are the only operation that creates genuinely new frequency content:** applying ReLU to a continuous signal can introduce arbitrarily high frequencies. They are therefore simultaneously the generator’s only source of new detail and its only source of aliasing. That observation is the paper’s intellectual core.

2.2 Equivariance metrics

Before changing anything the authors define two metrics, reported as PSNR in decibels. For translation,

$$\text{EQ-T} = 10 \log_{10} \left(I_{\max}^2 / \mathbb{E} \left[\left(g(t_x[z_0]; w)_c(p) - t_x[g(z_0; w)]_c(p) \right)^2 \right] \right), \quad (2)$$

comparing the generator’s output after translating its input against the translation of its output; EQ-R is the analogue for rotations drawn uniformly from $[0^\circ, 360^\circ)$. Higher is better. Defining the measurement first, then deriving architecture changes against it, is good practice and makes the subsequent ablation legible, though Section 4 argues it also creates a circularity the paper does not resolve.

2.3 Architectural changes

The redesign proceeds through named configurations, each isolating one change (Table 1).

B: Fourier features.

The learned 4×4 constant input is replaced by Fourier features, which define a spatially

infinite map that can be translated and rotated *exactly* rather than approximately. This is what makes the input side of the equivariance question well-posed at all.

C: No noise inputs.

StyleGAN2’s per-pixel noise is removed. It is a literal coordinate reference: stochastic detail keyed to grid position. Removing it costs the model its source of fine stochastic variation, a real sacrifice, discussed later.

D: Simplified generator.

Reduced mapping-network depth, and mixing and path-length regularisation disabled, to give the redesign a clean baseline.

E: Boundaries and upsampling.

Feature maps are given a small margin so the “infinite extent” assumption approximately holds near borders, and upsampling filters are improved.

F: Filtered nonlinearities.

The key change. The ideal operation is $\psi_s * \sigma(z)$: apply the nonlinearity in the continuous domain, then low-pass filter away the frequencies it created. This cannot be computed exactly, so it is approximated by **upsample** $\rightarrow$ **LReLU** $\rightarrow$ **downsample**, with a $2\times$ temporary resolution increase found sufficient in practice.

G: Non-critical sampling.

Filter cutoffs are lowered below the Nyquist limit, deliberately sacrificing a sliver of representable bandwidth to suppress the aliasing that critical sampling leaves behind.

H: Transformed Fourier features.

A learned global transform is applied to the input features, letting the model place the whole synthesis frame.

T: Flexible layer specifications.

Per-layer cutoffs and sampling rates become free parameters chosen by an explicit schedule rather than inherited from StyleGAN2. Yields **StyleGAN3-T**.

R: Rotation equivariance.

Separable filters are replaced by radially symmetric jinc filters and 3×3 convolutions by 1×1 , since neither separability nor square kernels are rotation-invariant. Yields **StyleGAN3-R**, with 56 % fewer parameters per layer.

2.4 Analysis

Strengths. The work is *theory-first* in a literature that is overwhelmingly empirical. The changes are not hyperparameter search; each is derived from a stated requirement, and the ablation shows each one paying off in the predicted direction. The identification of nonlinearities as the unique aliasing source is genuinely illuminating and transfers beyond GANs: any generative decoder with pointwise nonlinearities has the same defect.

Theory versus implementation. The continuous-domain argument is exact, but every realisation is an approximation, and the paper is candid that the approximations are empirical. The ideal filter has infinite support; a windowed Kaiser filter of size $n = 6$ is used. The ideal nonlinearity requires the continuous domain; $2\times$ upsampling is used because it “has been found sufficient”. The parameter ablation makes the gap visible: raising upsampling to $m = 4$ improves EQ-T from 66.65 to 74.21 dB but costs $2.31\times$ training time. Equivariance is therefore not achieved but *purchased*, along a continuum, and the operating point is chosen by taste rather than derived.

Treatment of the discriminator. Only the generator is treated. The discriminator remains an ordinary aliasing CNN, and it is the discriminator that defines the training signal. If D prefers to see front teeth at particular pixel locations, G is rewarded for putting them there, no matter how equivariant G ’s internals are. The paper notes exactly this in its limitations, and stops. Given that the same continuous-signal treatment would apply to D essentially unchanged, this is the most conspicuous piece of unfinished work in the paper.

3 Results and Outcomes

3.1 Ablation across configurations

Table 1: Progressive results on unaligned FFHQ at 256² (paper’s Figure 3, left). FID lower is better; EQ-T and EQ-R in dB, higher is better.

	Configuration	FID ↓	EQ-T ↑	EQ-R ↑
A	StyleGAN2	5.14	n/a	n/a
B	+ Fourier features	4.79	16.23	10.81
C	+ No noise inputs	4.54	15.81	10.84
D	+ Simplified generator	5.21	19.47	10.41
E	+ Boundaries & upsampling	6.02	24.62	10.97
F	+ Filtered nonlinearities	6.35	30.60	10.81
G	+ Non-critical sampling	4.78	43.90	10.84
H	+ Transformed Fourier features	4.64	45.20	10.61
T	+ Flexible layers (StyleGAN3-T)	4.62	63.01	13.12
R	+ Rotation equiv. (StyleGAN3-R)	4.50	66.65	40.48

Translation equivariance rises from 16.2 dB to 66.7 dB (roughly a 10⁵-fold reduction in equivariance error) while FID ends slightly *better* than the StyleGAN2 baseline (4.50 vs 5.14). Rotation equivariance is essentially untouched (~ 10.8 dB) until config R, which lifts it to 40.5 dB; this is informative, confirming that translation and rotation equivariance are separate properties requiring separate architectural treatment.

Two rows deserve attention as evidence of honest reporting. Configs E and F *degrade* FID (5.21 $\rightarrow$ 6.02 $\rightarrow$ 6.35) while improving EQ-T. The authors keep them because the theory demands them, and the loss is recovered at config G. A paper optimising for a headline number would have reordered or hidden this.

3.2 Generalisation across datasets

The pattern is consistent: EQ-T lands in the 59–65 dB range everywhere against 13–19 dB for StyleGAN2, while FID moves by small amounts in both directions. StyleGAN3 wins FID on three of the five sets shown and loses on aligned FFHQ (2.70 $\rightarrow$ 2.79/3.07), unsurprising, since aligned FFHQ is the dataset where a fixed coordinate frame is genuinely informative and coordinate dependence is therefore an advantage rather than a bug.

3.3 Cost

Equivariance is not free. StyleGAN3-R uses *half* the parameters of StyleGAN2 yet takes twice as long to train, because the filtered nonlinearities and radially symmetric filters are computationally heavy. The whole project consumed 92 GPU-years and 225 MWh.

Table 2: Results across six datasets (paper’s Figure 5, left). ADA [6] used for the smaller sets.

Dataset	Model	FID ↓	EQ-T ↑	EQ-R ↑
FFHQ-U 1024 ²	StyleGAN2	3.79	15.89	10.79
	StyleGAN3-T	3.67	61.69	13.95
	StyleGAN3-R	3.66	64.78	47.64
FFHQ 1024 ²	StyleGAN2	2.70	13.58	10.22
	StyleGAN3-T	2.79	61.21	13.82
	StyleGAN3-R	3.07	64.76	46.62
MetFaces 1024 ²	StyleGAN2	15.22	16.39	12.89
	StyleGAN3-T	15.11	65.23	16.82
	StyleGAN3-R	15.33	64.86	46.81
AFHQv2 512 ²	StyleGAN2	4.62	13.83	11.50
	StyleGAN3-T	4.04	60.15	13.51
	StyleGAN3-R	4.40	64.89	40.34
Beaches 512 ²	StyleGAN2	5.03	15.73	12.69
	StyleGAN3-T	4.32	59.33	15.88
	StyleGAN3-R	4.57	63.66	37.42

Table 3: Training cost on FFHQ 1024² (paper, Section 4).

Model	Parameters	GPU-hours	Overhead
StyleGAN2	30.0M	1106	n/a
StyleGAN3-T	22.3M	1576	+42 %
StyleGAN3-R	15.8M	2248	+103 %

3.4 Ablation studies

Several ablations are more informative than the headline numbers:

- **Re-enabling noise inputs collapses equivariance:** EQ-T drops 63.01 $\rightarrow$ 24.46 dB. Direct confirmation that per-pixel noise is a coordinate channel, not innocuous stochasticity.
- **Path-length regularisation harms FID** (4.62 $\rightarrow$ 5.00) while *improving* EQ-T. The authors reason that it penalises image change under latent motion and so actively encourages sticking, and suspect the equivariance gain comes from blurrier output, a candid, self-undermining observation.
- **Lanczos filters damage rotation equivariance** (EQ-R 40.48 $\rightarrow$ 25.25) because separable filters are not radially symmetric, precisely as the theory predicts.
- **A p4 G-CNN alternative** [1] reaches only 17.07 dB EQ-R at 2.48 $\times$ the training time, against 40.48 dB for config R. Steerable filters [11] were found infeasible at this network size.
- **Internal representations change qualitatively:** some feature maps switch from encoding magnitude to encoding *phase*, which is what a network must do to track sub-pixel position.

3.5 Scope of the evidence

Within its declared scope the evidence is thorough and honest. The scope, however, is narrower than the abstract’s closing claim that the results “pave the way for generative models better suited for video and animation”. No video model is trained; no temporal metric such as FVD [10] is reported; the video evidence is a set of qualitative accompanying clips. The quantitative case rests entirely on two metrics the paper itself introduces.

4 Limitations and Research Gaps

4.1 Limitations acknowledged by the authors

1. **The discriminator is untreated.** Conceded to cause visible failures: FFHQ teeth do not move correctly when the head turns.
2. **Assumptions about the data can be violated.** If the training data is itself aliased (point-sampled cartoons, letterboxing, low-quality JPEG, retro pixel art), then jagged, grid-locked edges are a *defining* feature to be reproduced, and smooth sub-pixel translation directly contradicts fidelity to the data.
3. **Stochastic variation was removed, not replaced.** Noise inputs had to go; nothing consistent with hierarchical synthesis was put in their place.
4. **Only rigid motions are handled.** Scaling, anisotropic scaling and general deformations are listed as future work.
5. **Everything happens in sRGB.** Antialiasing should precede tone mapping; all GANs including this one operate after it.
6. **Attention- and token-based generators may be incompatible.** VQGAN-style tokenisers [2] are flagged as “possibly at odds” with equivariance, and whether they can be made equivariant is left open.
7. **Cost.** 92 GPU-years and 225 MWh for the project; +103 % training time for the rotation-equivariant configuration.

4.2 Additional gaps

G1. Self-validating metrics. EQ-T and EQ-R are introduced by the same paper that optimises them, and no external task confirms that more decibels means a better generator. The paper’s own guidance (that EQ-T above ~ 50 dB and EQ-R above ~ 40 dB “look good”) is calibrated by watching videos. This is a closed loop: the architecture is designed to satisfy a property, the property is measured by a purpose-built metric, and the metric’s threshold is set by inspecting outputs of that architecture. Nothing here is dishonest, but nothing independently anchors the scale either.

G2. FID parity is a non-regression. “Matches the FID of StyleGAN2” means image quality did not get worse. The entire positive value proposition therefore lives in video and animation, which is exactly the setting that receives no quantitative evaluation. The paper’s strongest claim and its weakest evidence coincide.

G3. Global equivariance versus local motion. The opening example is *hierarchical* and *local*: moving a head moves the nose, which moves the pores. What the method achieves and measures is equivariance to *global* rigid transformation of the entire image. A generator can be perfectly equivariant under global translation while still failing to move a nose correctly with a turning head, because that is a local, non-rigid deformation. The FFHQ teeth failure is precisely an instance of this, and attributing it solely to the discriminator may be too generous to the method.

G4. Equivariance assumed universally desirable. The paper treats coordinate dependence as pathology. But its own aligned-FFHQ result shows StyleGAN2 winning on FID where

the coordinate frame is meaningful, and its own cartoon example concedes that grid-locked structure is sometimes correct. There is no mechanism for a model to be equivariant *where appropriate* and coordinate-dependent elsewhere; the choice is made globally, at architecture design time, by the practitioner.

G5. Scientific and medical imaging. This gap follows from G4 and is the one I find most consequential. In B-mode ultrasound (the modality of my course project) image texture is *speckle*: a coherent interference pattern determined by the transducer geometry and the beam-forming grid, not by the surfaces of the tissue. Speckle genuinely is “stuck” to the imaging coordinate frame, and a physically faithful generator of ultrasound *should* keep it there while allowing anatomy to move independently. StyleGAN3’s architecture forbids exactly this. The same argument applies to CT streak artefacts, MRI coil-sensitivity shading, and the burnt-in depth rulers and on-screen chrome present in essentially every clinical ultrasound frame.

G6. Limited domain coverage. Five of the six datasets are natural images dominated by a single centred subject. Neither a texture-dominated domain nor a structured scientific domain is tested, so the claim of “generally applicable, small architectural changes” is broader than what was shown.

5 Concluding Remarks and Suggested Improvements

The reframing at this paper’s heart (treat feature maps as bandlimited continuous signals, and every layer as an operator on them) is correct, useful and transferable, and I expect it to outlive the specific architecture. My suggestions below are aimed at the gaps of Section 4, ordered from cheapest to most speculative.

S1. Equivariant discriminator

The authors attribute the FFHQ teeth failure to a coordinate-dependent discriminator but do not test it. The fix needs no new theory: the filtered-nonlinearity construction (upsample $\rightarrow$ LReLU $\rightarrow$ downsample) and the non-critical sampling schedule apply to D unchanged. **Concretely**, train three models on FFHQ: equivariant G only (the paper), equivariant D only, and both, and report EQ-T alongside a targeted measurement of the teeth artefact (landmark displacement of the dental region across a head-turn interpolation). This is a falsifiable test of the paper’s own hypothesis, costs one additional training run per arm, and would either close the most obvious gap or show that the explanation was wrong. Either outcome is publishable.

S2. Independent validation of equivariance

To break the closed loop of G1, equivariance should be validated by a tool that knows nothing about it. **Concretely**, generate latent-interpolation sequences and measure temporal coherence with an *independent* optical-flow estimator such as RAFT [9]: for a genuinely equivariant generator, the flow field between consecutive frames should be smooth and should integrate consistently over the sequence, whereas texture sticking produces near-zero flow in textured regions while the underlying structure moves, a signature that is easy to quantify as the divergence between flow measured on structure and on texture. Reporting FVD [10] on generated sequences would similarly convert the abstract’s video claim into evidence.

I weight this suggestion heavily because of an experience in my own project: I validated a derived orientation target using a structure tensor, then found the number only trustworthy once I

checked it against an entirely independent principal-axis estimator. Self-consistent measurements are comfortable and can be confidently wrong.

S3. Adaptive band-limiting

The paper applies the same expensive treatment at every layer, and its own parameter ablation shows a clear Pareto frontier that it does not exploit: filter size $n = 4$ gives 57.5 dB at $0.84\times$ time, $n = 6$ gives 66.7 dB at $1.00\times$, upsampling $m = 4$ gives 74.2 dB at $2.31\times$. Since aliasing is created only at nonlinearities and only where the signal actually contains near-Nyquist energy, the filtering effort should follow the spectrum. **Concretely**, measure per-layer spectral occupancy during training and set each layer’s upsampling factor and filter support from it, spending $m = 4$ only in the few layers that need it. Given StyleGAN3-R’s +103 % overhead, recovering even half of that at equal EQ-R would materially change the method’s practicality.

S4. Local deformation equivariance

This addresses G3, the gap between what the paper motivates and what it delivers. Global translation and rotation are the trivial members of the transformation family the introduction actually describes. **Concretely**, extend the input Fourier-feature grid to accept a smooth, low-frequency *deformation field* rather than a rigid transform, and define EQ-D by analogy with EQ-T using small-amplitude random warps. The continuous-signal argument carries over directly, since a smooth warp of a bandlimited signal is still bandlimited for sufficiently small amplitude; the bandwidth budget simply has to account for the warp’s Jacobian. This would let one state and test the paper’s motivating claim (that a nose moves with a head) rather than a proxy for it. Spatial transformer machinery [5] offers a ready parameterisation.

S5. Selective equivariance

Gaps G4 and G5 share a root cause: equivariance is imposed globally when it is really a property of *content*. Some image content belongs to object surfaces and should move with them; other content belongs to the imaging process and should not. **Concretely**, I propose a *two-stream generator*: an equivariant structural stream built exactly as StyleGAN3 prescribes, and a deliberately coordinate-dependent stream that retains per-pixel noise and unfiltered positional encoding, with a learned spatial gate combining them. The gate is supervised by nothing; it is free to discover which content is surface-bound.

This also restores what config C threw away. StyleGAN3 removed noise inputs because they leak coordinates; the two-stream design keeps them where leaking coordinates is physically correct.

The test case I would use is ultrasound synthesis. In B-mode imaging, anatomy (aponeuroses, fascicles, muscle boundaries) is surface-bound and should be equivariant, while speckle is beam-bound and should not be, and the on-screen depth ruler and probe annotations are rigidly fixed to the display frame. A single generator currently cannot represent that split. A successful two-stream model would be checkable objectively rather than by eye: measure whether the synthesised speckle retains the correct first-order statistics (Rayleigh distributed under fully developed speckle) while anatomical structures translate correctly. Synthetic ultrasound with realistic speckle would be genuinely useful, since annotated medical data is scarce; my own project had 1 048 annotated aponeurosis frames in total [8].

S6. Layer-selective rotation equivariance

StyleGAN3-R doubles training time to make *every* layer rotation-equivariant. But the paper’s own analysis holds that coarse layers determine structure while fine layers add detail, and its config-T contribution made per-layer specifications a free parameter. **Concretely**, build a hybrid: radially symmetric jinc filters and 1×1 convolutions in the low-resolution layers where global pose is decided, and cheaper translation-only treatment at high resolution. If most of the 40.5 dB EQ-R survives at closer to StyleGAN3-T’s +42 % overhead rather than +103 %, rotation equivariance becomes affordable by default instead of a deliberate trade.

Closing assessment

This is a strong paper whose durable contribution is diagnostic rather than architectural: it shows that a defect invisible to the field’s dominant metric had been present in every state-of-the-art generator, and it supplies the signal-processing vocabulary to reason about such defects. The architecture that follows is a competent, expensive, and somewhat rigid instantiation of that idea. Its principal unexamined assumption (that dependence on image coordinates is always pathological) happens to hold for photographs of faces and animals, and to fail for a large class of scientific and medical imagery where the coordinate frame is physically meaningful. The most valuable extension is therefore not more equivariance but *selective* equivariance: a generator that learns which of its content belongs to the world and which belongs to the camera.

References

- [1] Taco S. Cohen and Max Welling. Group equivariant convolutional networks. In *International Conference on Machine Learning (ICML)*, pages 2990–2999, 2016.
- [2] Patrick Esser, Robin Rombach, and Björn Ommer. Taming transformers for high-resolution image synthesis. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 12873–12883, 2021.
- [3] Ian Goodfellow, Jean Pouget-Abadie, Mehdi Mirza, Bing Xu, David Warde-Farley, Sherjil Ozair, Aaron Courville, and Yoshua Bengio. Generative adversarial nets. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2014.
- [4] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.
- [5] Max Jaderberg, Karen Simonyan, Andrew Zisserman, and Koray Kavukcuoglu. Spatial transformer networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2015.
- [6] Tero Karras, Miika Aittala, Janne Hellsten, Samuli Laine, Jaakko Lehtinen, and Timo Aila. Training generative adversarial networks with limited data. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2020.
- [7] Tero Karras, Samuli Laine, Miika Aittala, Janne Hellsten, Jaakko Lehtinen, and Timo Aila. Analyzing and improving the image quality of StyleGAN. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 8110–8119, 2020.
- [8] Paul Ritsche, Martino V. Franchi, Oliver Faude, Taija Finni, Olivier Seynnes, and Neil J. Cronin. Fully automated analysis of muscle architecture from B-mode ultrasound images with DL_Track_US. *Ultrasound in Medicine & Biology*, 50(2):258–267, 2024.

- [9] Zachary Teed and Jia Deng. RAFT: Recurrent all-pairs field transforms for optical flow. In *European Conference on Computer Vision (ECCV)*, pages 402–419, 2020.
- [10] Thomas Unterthiner, Sjoerd van Steenkiste, Karol Kurach, Raphaël Marinier, Marcin Michalski, and Sylvain Gelly. Towards accurate generative models of video: A new metric and challenges. In *arXiv:1812.01717*, 2018.
- [11] Maurice Weiler and Gabriele Cesa. General $E(2)$ -equivariant steerable CNNs. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.